PARKINOVA

Medical Assessment Report

A. Patient & Assessment Information

Patient Reference	arun
Age / Gender	20 / Male
Assessment Date	2026-11-01 04:25:55
Report Reference ID	89e909ab-52c6-47e2-939c-5262b69c86ae
Completed Modules	Brain MRI Scan, Archimedean Spiral Analysis, Symptom Questionnaire
Lead Evaluator	Dr. Researcher

B. Combined Assessment Summary

Analysis Component	Completion Status	Model Prediction
Brain MRI Scan	Completed	No Parkinson's-related pattern detected
Spiral Drawing	Completed (Model Prediction)	Spiral drawing shows motor irregularities correlated with Parkinsonian tremor patterns

Multi-Modal Synthesis: Divergent Findings (Disagreement Across Modalities)

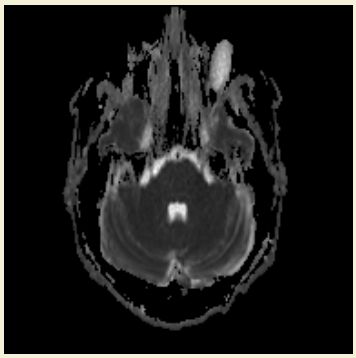
The MRI scan analysis and spiral drawing analysis yielded divergent results. In neurology, structural imaging and motor performance frequently diverge: early structural changes can occur prior to tremor, or physiological tremor may affect drawing sketches while 3T MRI is unremarkable. Both results are presented transparently; direct clinical evaluation by a licensed neurologist is essential.

C. Patient Symptom & Cognitive Assessment Profile

Symptom Domain	Category	Patient Response
Memory Difficulties	*Cognitive*	Rarely
Difficulty Concentrating	*Cognitive*	Sometimes
Problems with Decision-Making	*Cognitive*	Rarely
Changes in Mood	*Emotional*	Sometimes
Feelings of Anxiety	*Emotional*	Rarely
Symptoms of Depression	*Emotional*	Often
Self-Reported Emotional Well-being	*Emotional*	Sometimes
Sleep Disturbances	*Sleep & Energy*	Rarely
Mental Fatigue	*Sleep & Energy*	Sometimes
Reduced Motivation	*Sleep & Energy*	Sometimes
Difficulty with Daily Activities	*Functional*	Rarely
Changes in Thinking Speed	*Functional*	Sometimes
Difficulty Planning & Organizing	*Functional*	Rarely

Note: 13 of 13 self-reported questions completed on 2026-11-01 09:55:36. Symptom frequencies are self-reported for research and risk documentation purposes only; they do not independently constitute a diagnostic evaluation.

D. Brain MRI Scan Analysis

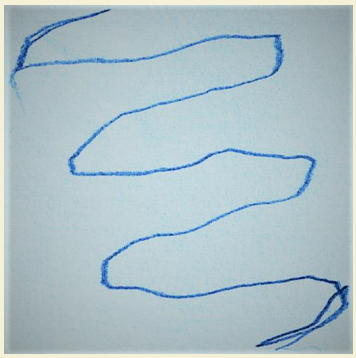


Analysis Status	Completed
Model Prediction	No Parkinson's-related pattern detected
Model Architecture	Parkinson CNN (244x244x3)
Input Preprocessing	244x244 Lanczos Resizing, Channel Standardization
Image Source	ep2d_diff_3scan_trace_p2_ADC_DFC_009.png

Model Findings & Interpretation:

The CNN deep learning model did not detect distinctive Parkinson's-associated patterns in the analyzed MRI scan compared to the reference control dataset. This experimental finding does not rule out early-stage disease, atypical parkinsonism, or other neurological conditions.

E. Archimedean Spiral Drawing Analysis



Analysis Status	Completed (Model Prediction)
Model Prediction	Spiral drawing shows motor irregularities correlated with Parkinsonian tremor patterns
Model Architecture	Archimedean Spiral YOLO / CNN (244x244x3)
Geometry & Stroke	Analyzed | Standard
Drawing Source	Parkinson954.png

Visual & Kinematic Observations:

- **dimensions:** 516 x 512 pixels
- **aspect_ratio:** 1.008:1
- **stroke_density_pct:** 10.13
- **canvas_style:** Light background (paper / digital canvas)
- **contrast_range:** 225.7
- **file_size_kb:** 26.2
- **Model Status:** YOLO Model Active (best)
- **Input Resolution:** 224x224 px
- **Predicted Class:** Parkinson

Regulatory Notice & Academic Disclaimer

This application is an academic AI research prototype. Its output is generated by a machine-learning model and is not a medical diagnosis. The model's performance and reliability may vary depending on image quality, dataset characteristics, and the images used during training. Consult a qualified neurologist for medical evaluation.